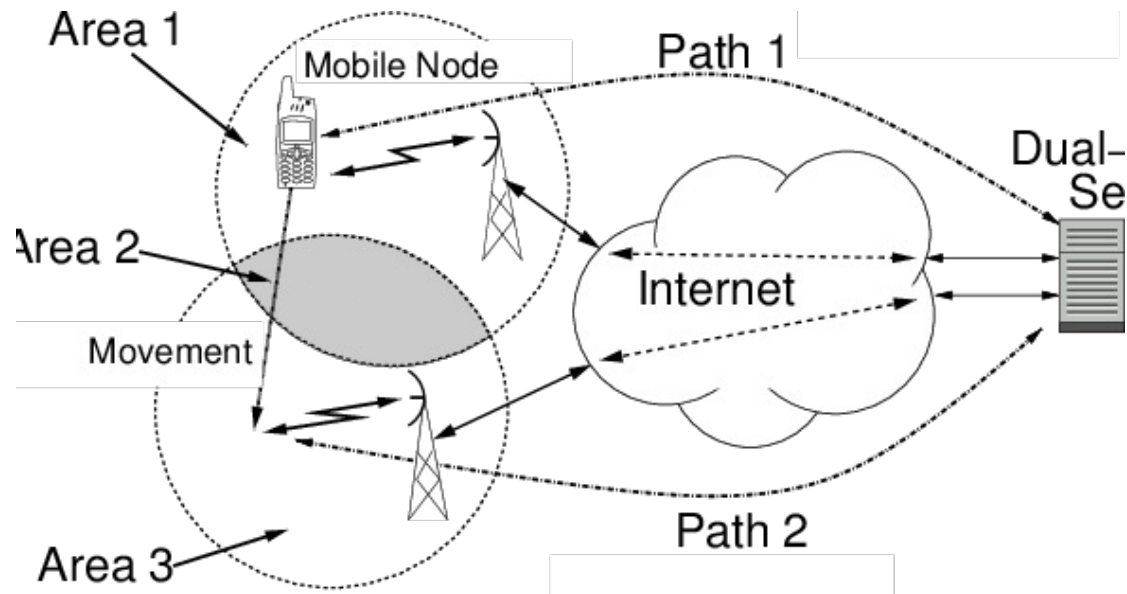


Mobile IP

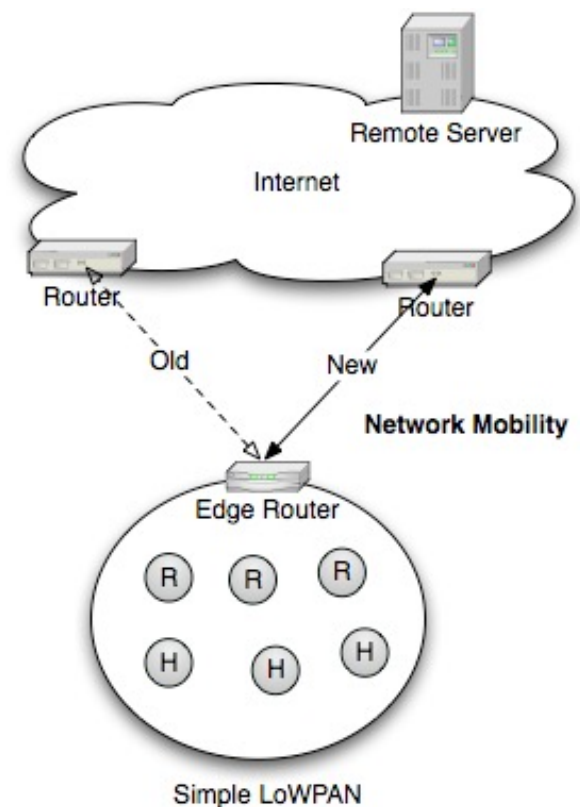
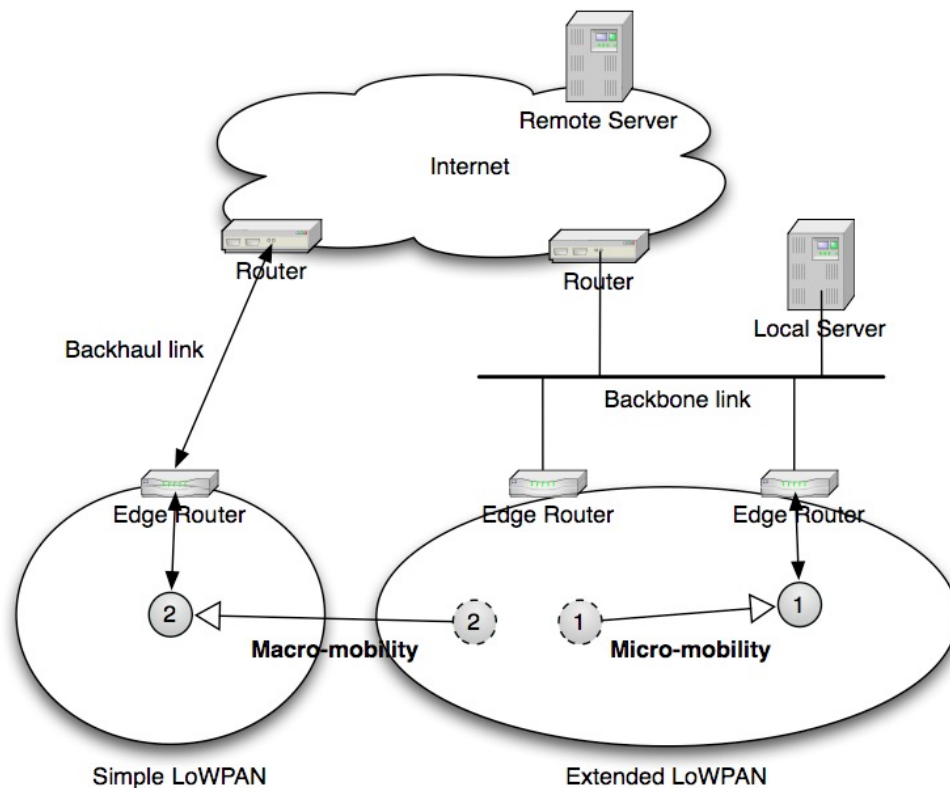
Mobility

- What causes mobility?
 - Physical movement
 - Radio channel availability changes
 - Network performance
 - Sleep schedules
 - Node failure



Types of Mobility

- Mobility can be categorized as
 - Micro-mobility - *within a network domain*
 - Macro-mobility - *between network domains (IP address change)*
- *We can experience Node or Network mobility*



Effect of mobility on protocol stack

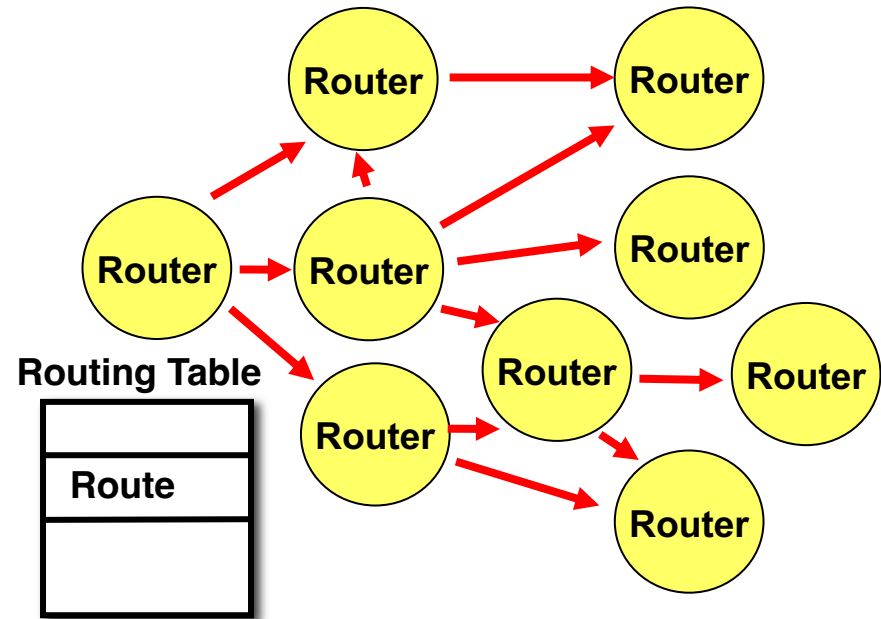
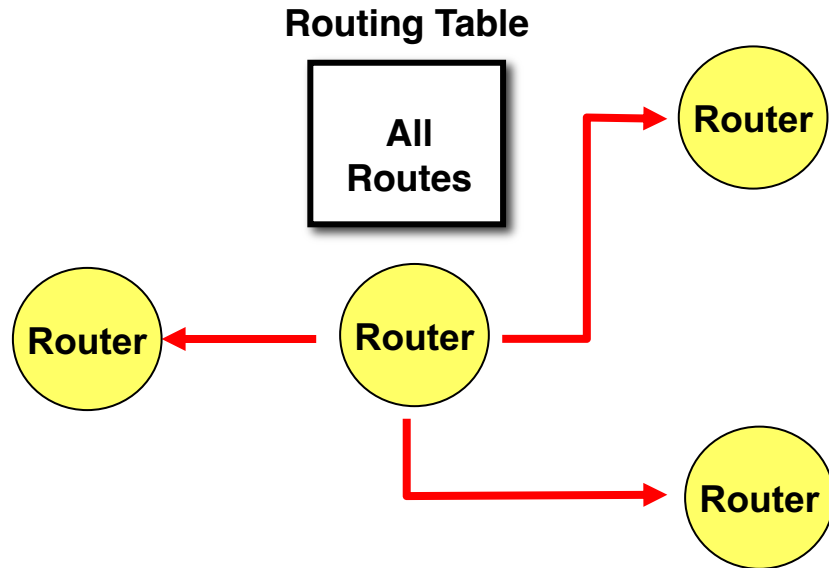
- Application
 - new applications and adaptations
 - service location, multimedia
- Transport
 - congestion and flow control
 - quality of service
- Network
 - addressing and routing
 - device location, hand-over
- Link
 - media access and security
- Physical
 - transmission errors and interference

Perspectives

- **Network designers:** Concerned with cost-effective design
 - Need to ensure that network resources are efficiently utilized and fairly allocated to different users.
- **Network users:** Concerned with application services
 - Need guarantees that each message sent will be delivered without error within a certain amount of time.
- **Network providers:** Concerned with system administration
 - Need mechanisms for security, management, fault-tolerance and accounting.

Traditional Routing

- A *routing protocol* sets up a *routing table* in *routers*



- Routing protocols are typically based on Distance-Vector or Link-State schemes

Routing and Mobility

- Finding a path from a source to a destination
- Issues
 - Frequent route changes
 - amount of data transferred between route changes may be much smaller than traditional networks
 - Route changes may be related to host movement
 - Low bandwidth links
- Goal of routing protocols
 - decrease routing-related overhead
 - find short routes
 - find “stable” routes (despite mobility)

Roaming and Handover

- **Roaming:**

- A mobile node moves from one network to another, typically with no existing packet streams.

- **Handover:**

- A mobile node disconnects from its existing point of attachment (AP) within the same network and attaches itself to a new one
 - May include operations at the link and network layers as for the mobile node to be able to communicate again.
 - One or more application packet streams typically accompany the mobile node as it undergoes handover.

Mobile IP (RFC 3344): Motivation

- **Traditional routing**

- based on IP address; network prefix determines the subnet
- change of physical subnet implies
 - change of IP address (conform to new subnet), or
 - special routing table entries to forward packets to new subnet

- **Changing of IP address**

- DNS updates take to long time
- TCP connections break
- security problems

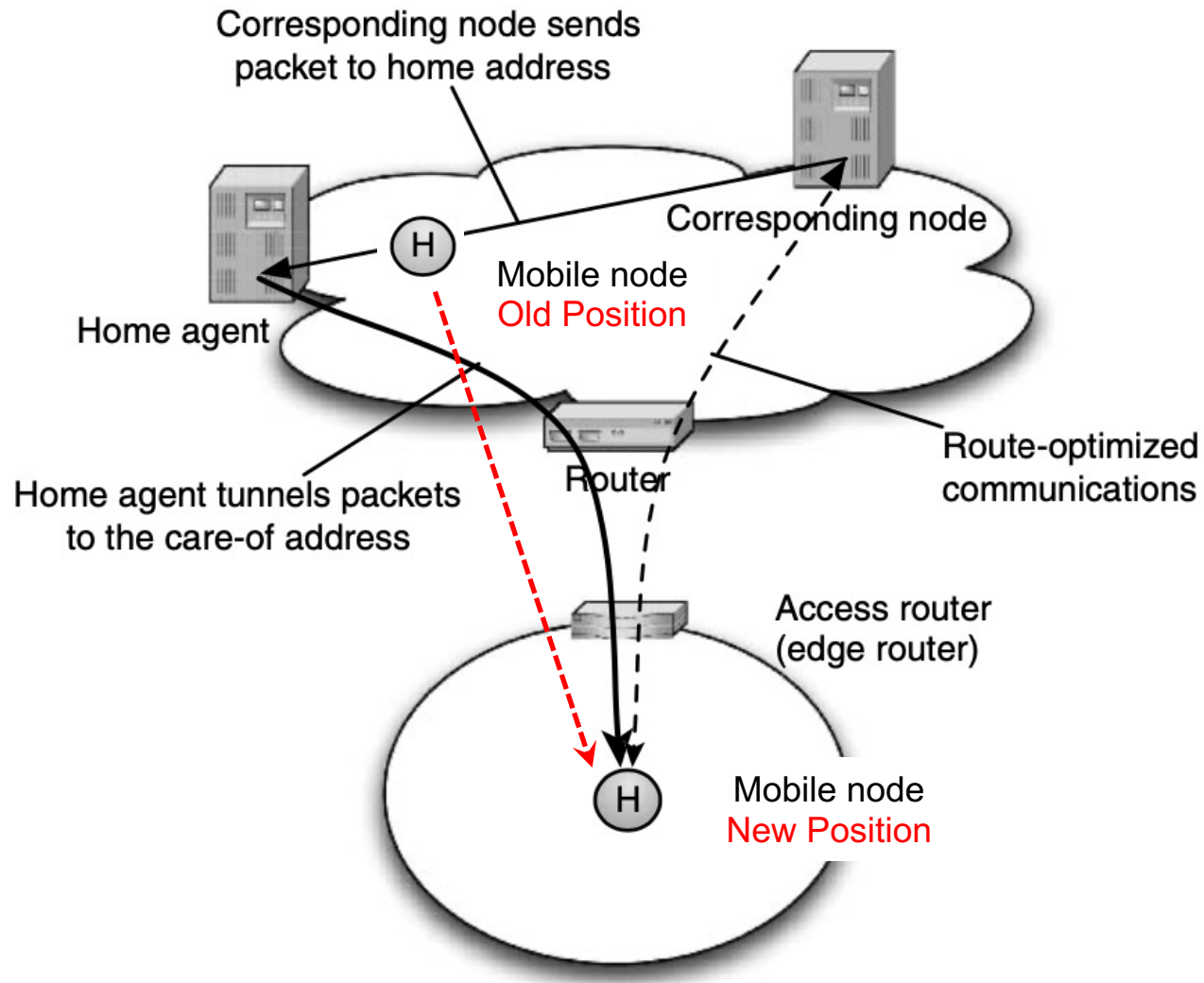
- **Changing entries in routing tables**

- does not scale with the number of mobile hosts and frequent changes in the location
- security problems

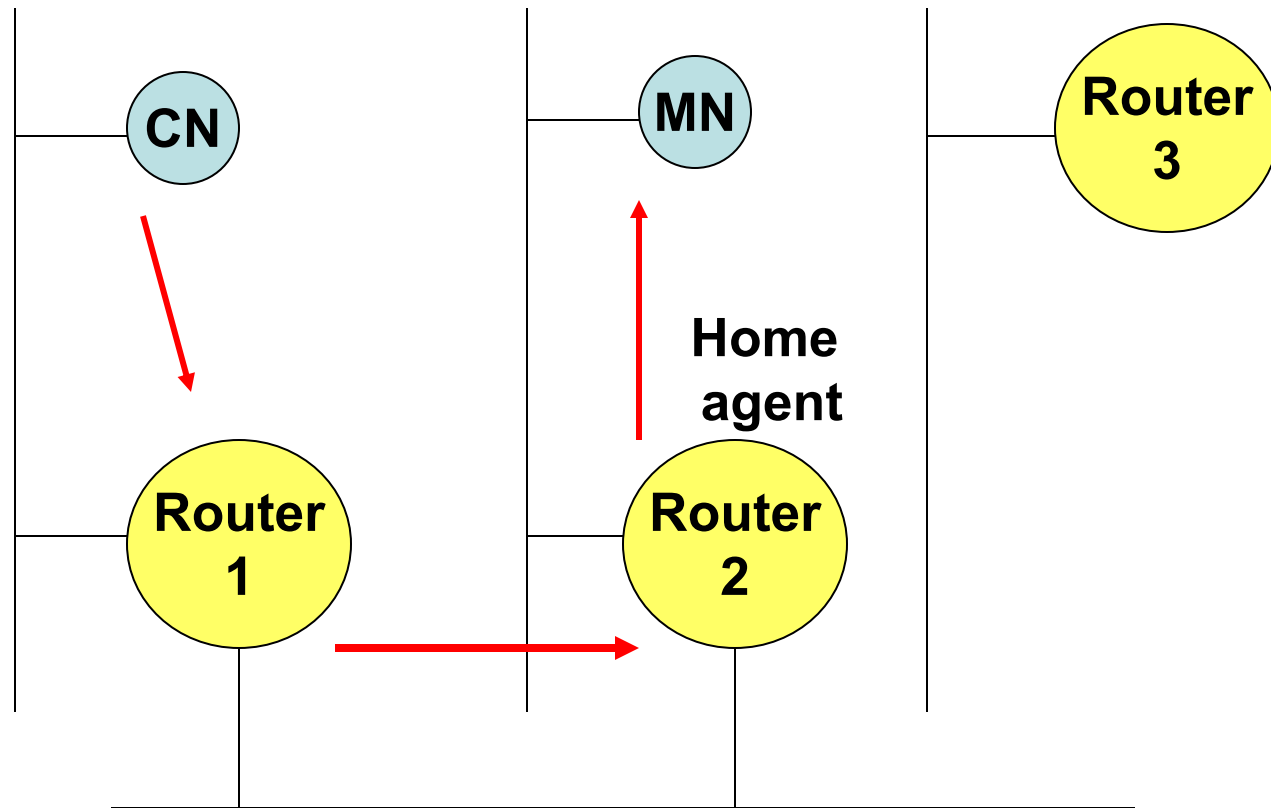
- **Solution requirements**

- retain same IP address, use same layer 2 protocols
- authentication of registration messages, ...

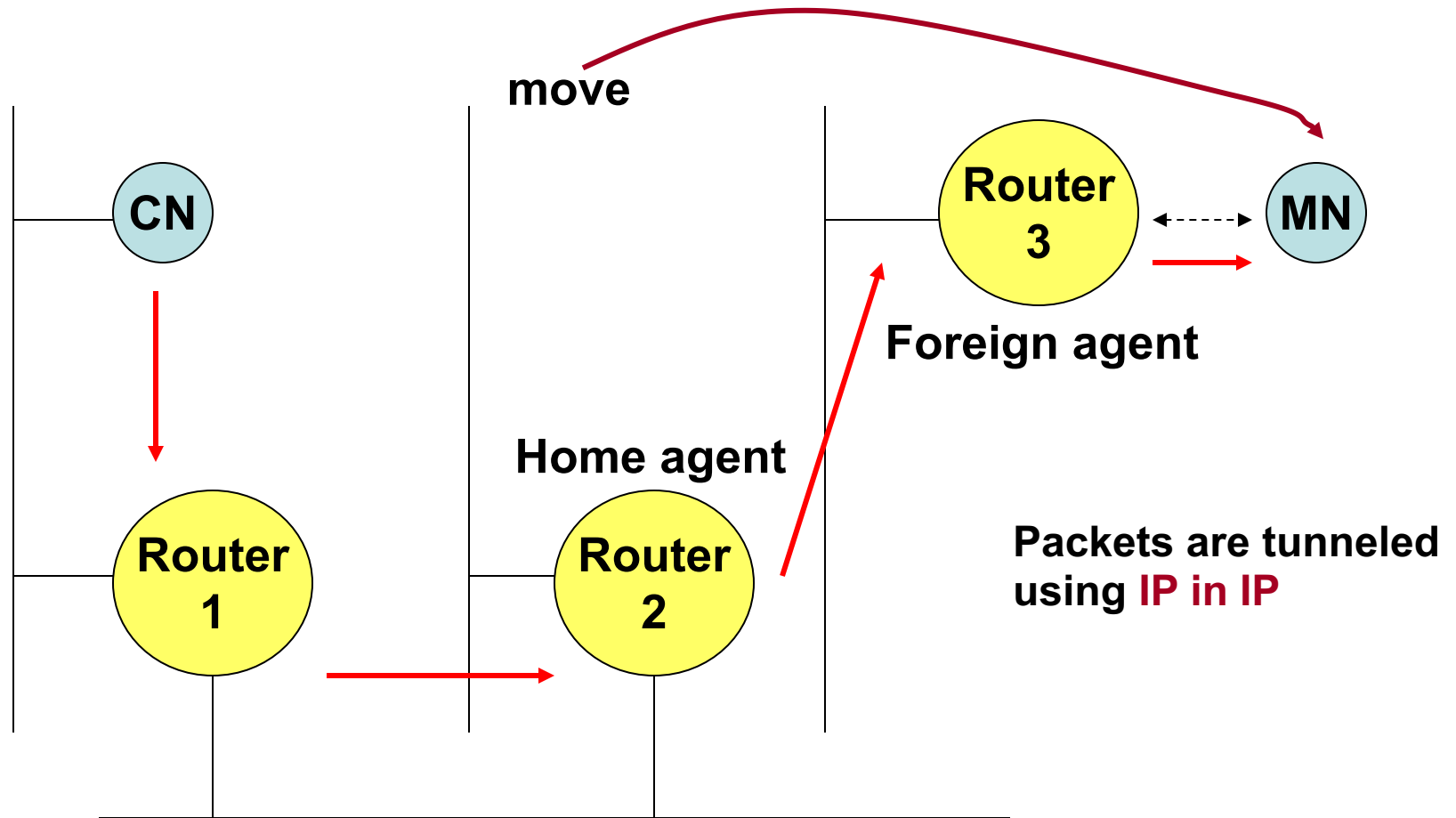
Mobile IP scenario



Mobile IP: Basic Idea



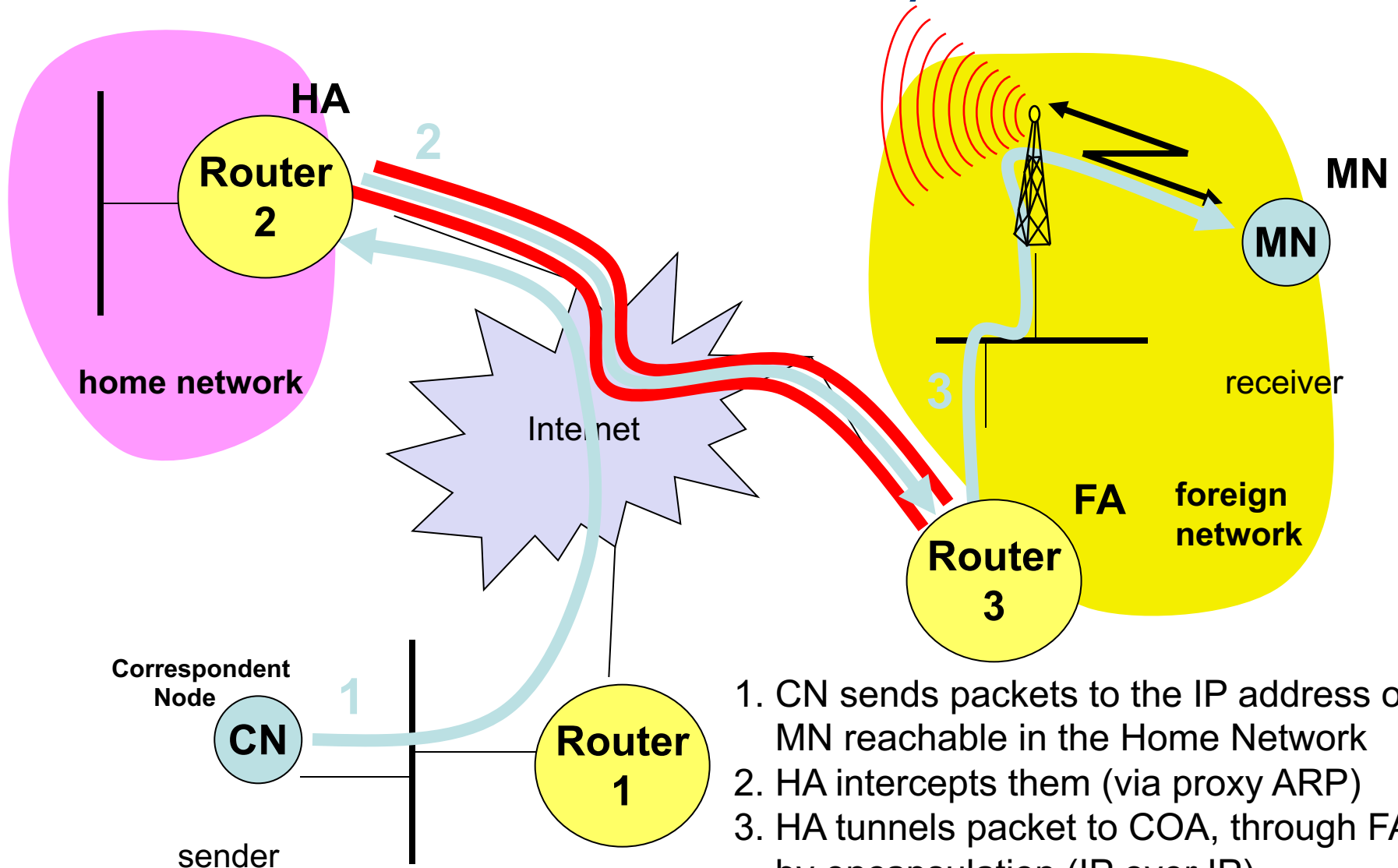
Mobile IP: Basic Idea



Mobile IP: Terminology

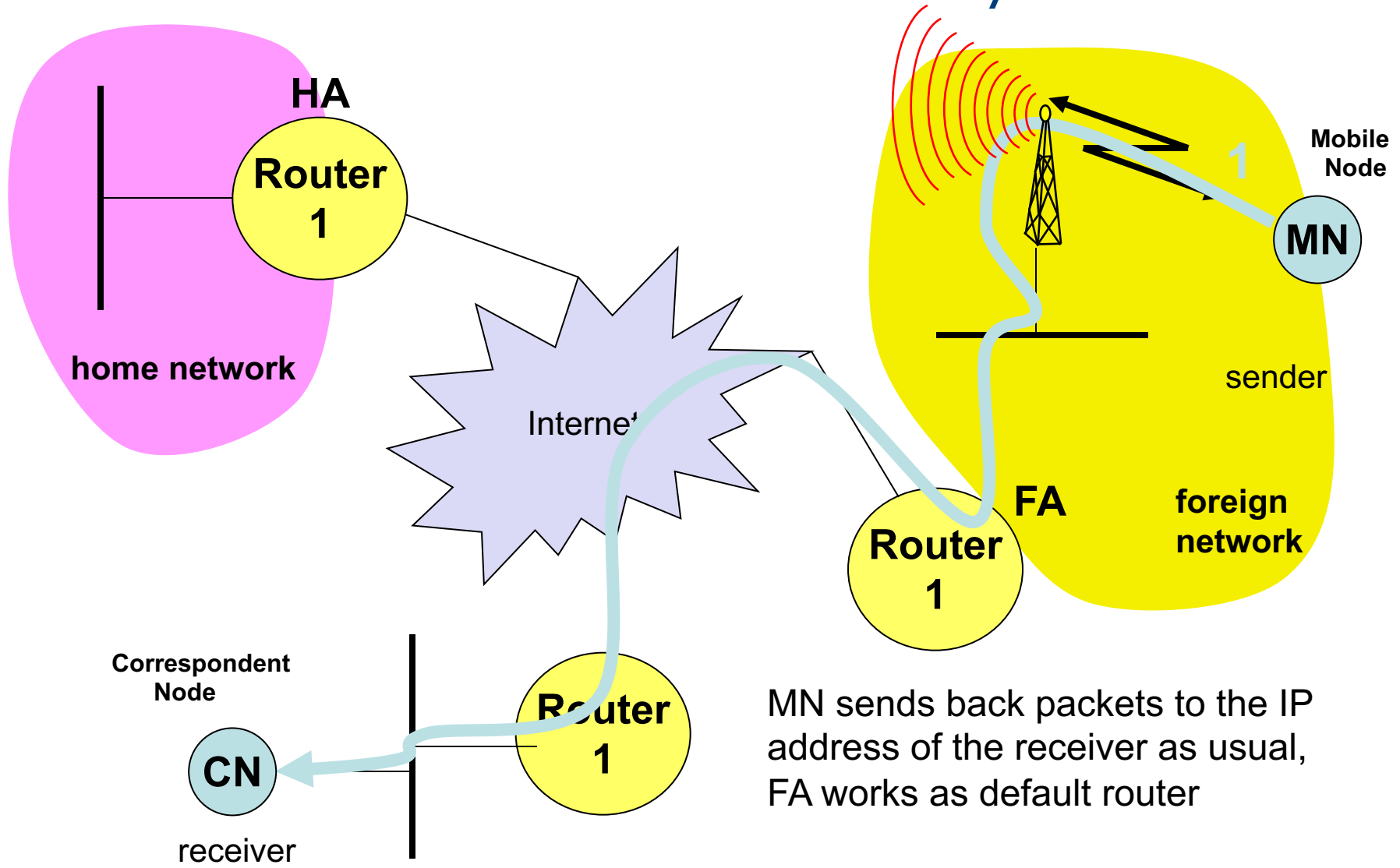
- **Mobile Node (MN)**
 - node that moves across networks without changing its IP address
- **Home Agent (HA)**
 - host in the home network of the MN, typically a router
 - registers the location of the MN, **tunnels IP packets** to the COA
- **Foreign Agent (FA)**
 - host in the current foreign network of the MN, typically a router
 - forwards tunneled packets to the MN, typically the default router for MN
- **Care-of Address (COA)**
 - address of the **current tunnel end-point** for the MN (at FA or MN)
 - actual location of the MN from an IP point of view
- **Correspondent Node (CN)**
 - host with which MN is “corresponding” (TCP connection)

Data transfer to the mobile system



1. CN sends packets to the IP address of MN reachable in the Home Network
2. HA intercepts them (via proxy ARP)
3. HA tunnels packet to COA, through FA, by encapsulation (IP over IP)
4. FA forwards the packet to the MN

Data transfer from the mobile system



Mobile IP: Basic Operation

– Agent Advertisement

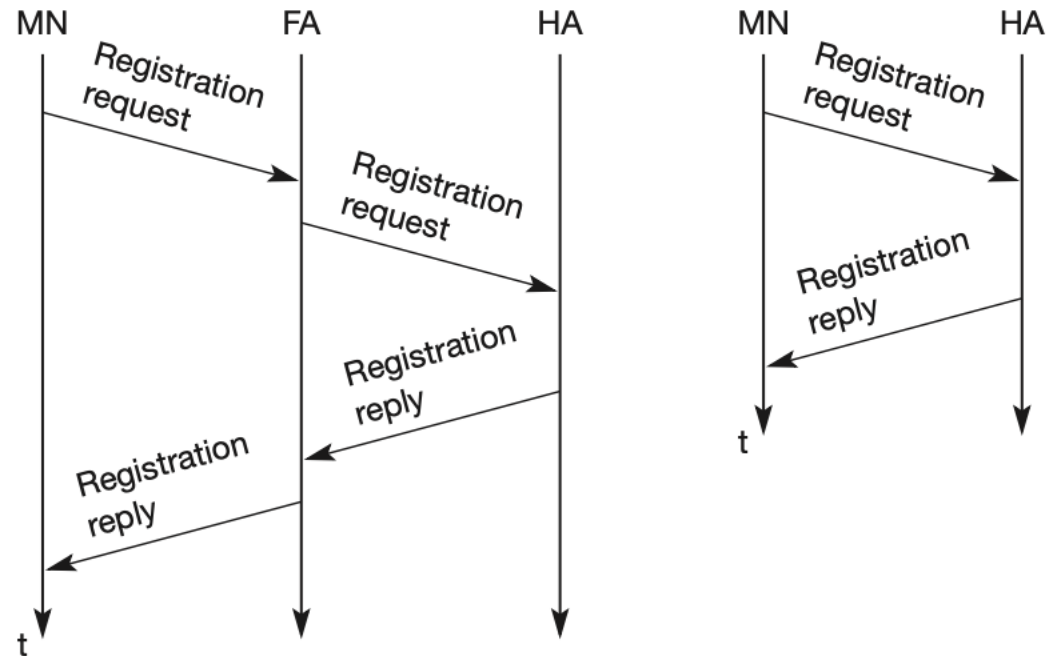
- HA/FA periodically send advertisement messages into their physical subnets
- MN listens to these messages and detects, if it is in home/foreign network
- MN reads a COA from the FA advertisement messages

0	7	8	15	16	23	24	31					
type		code		checksum								
#addresses		addr. size		lifetime								
router address 1												
preference level 1												
router address 2												
preference level 2												
...												
type = 16		length		sequence number								
registration lifetime				R	B	H	F	M	G	r	T	reserved
COA 1												
COA 2												
...												

Mobile IP: Basic Operation

– MN Registration

- MN signals COA to the HA via the FA
- HA acknowledges via FA to MN
- MN can also directly register in its home network
- limited lifetime, need to be secured by authentication



0	7	8					15	16		23	24	31
type 1		S	B	D	M	G	r	T	x	lifetime		
home address												
home agent												
COA												
identification												
extensions ...												

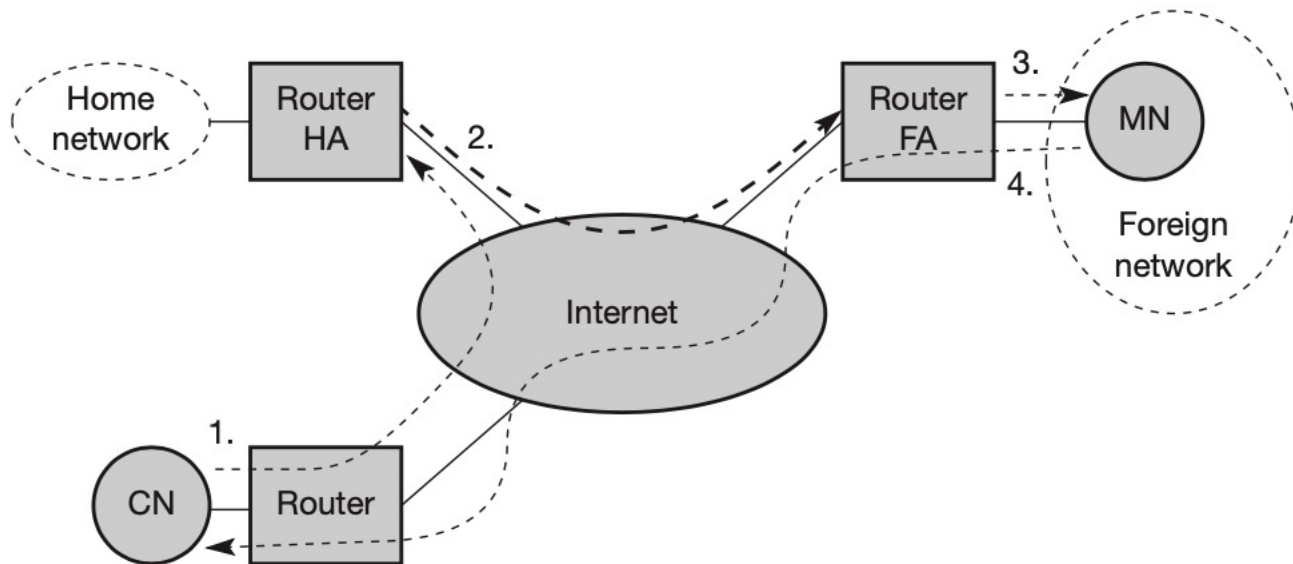
Mobile IP: Basic Operation

– HA Proxy

- HA advertises the IP address of the MN (as for fixed systems)
- packets to the MN are sent to the HA
- independent of changes in COA/FA

– Packet Tunneling

- HA to MN via FA



Discovering the care-of address

- Discovery process built on top of an existing standard protocol: router advertisement (RFC 1256)
- Router advertisements extended to carry available care-of addresses called resulting into **agent advertisements**
- Foreign agents (and home agents) send agent advertisements periodically
- A mobile host can choose not to wait for an advertisement, and issue a solicitation message

Agent advertisements

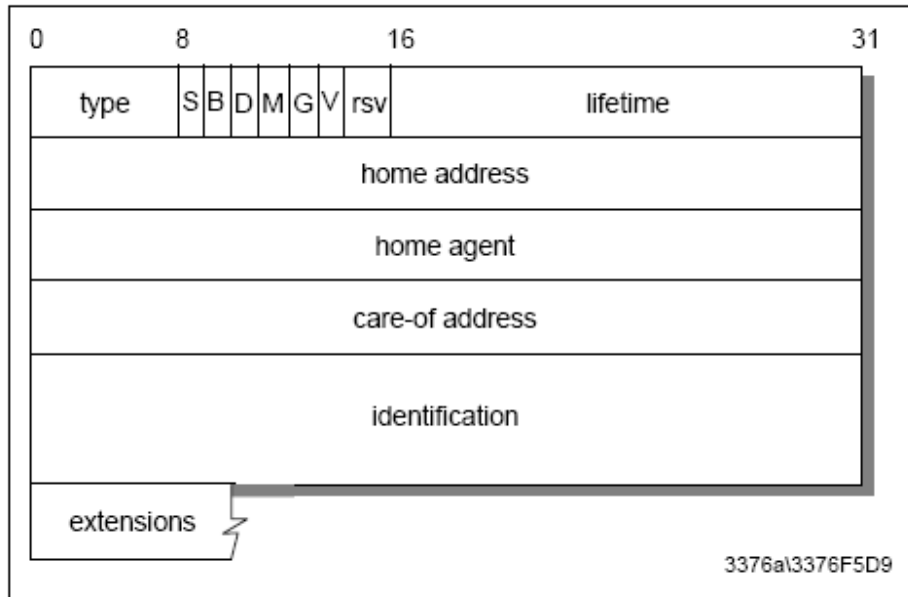
- Foreign agents send advertisements to advertise available care-of addresses
- Home agents send advertisements to make themselves known
- Mobile hosts can issue agent solicitations to actively seek information
- If mobile host has not heard from a foreign agent its current care-of address belongs to, it seeks for another care-of address

Registering the Care-of Address

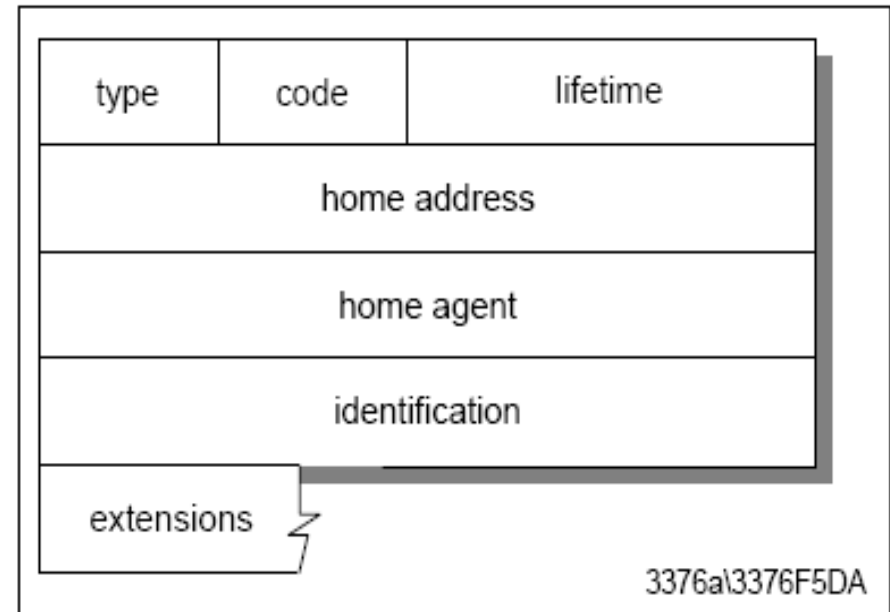
- Once mobile host receives care-of address, it registers it with the home agent
- A registration request is first sent to the home agent (through the foreign agent)
- Home agent then approves the request and sends a registration reply back to the mobile host
- Security?

Mobile IP Registration

1. Registration Request by MN to FA
2. FA Relays Registration request to HA
3. HA sends Registration reply to FA
4. FA Relays Registration reply to MN

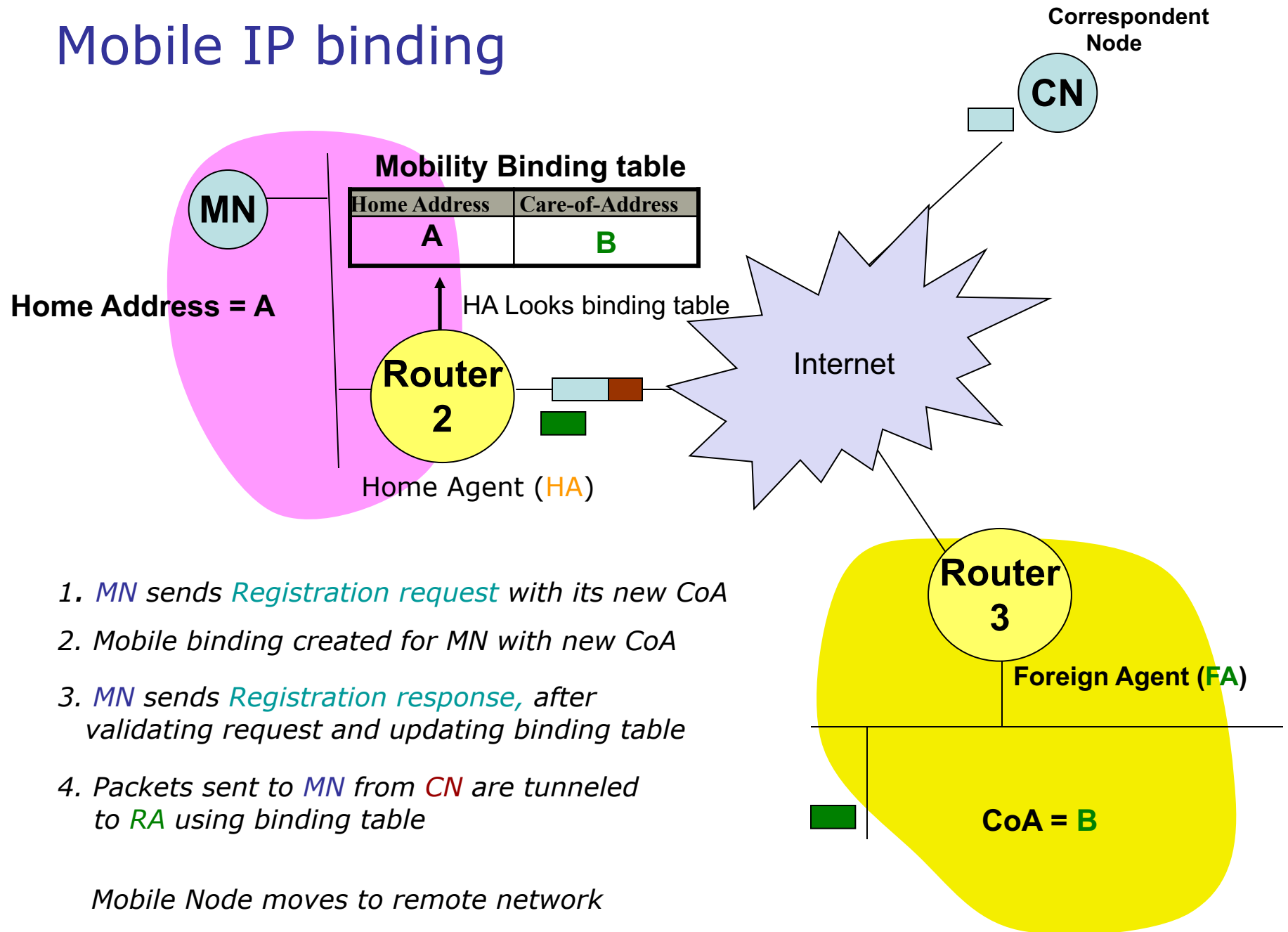


Register request



Register response

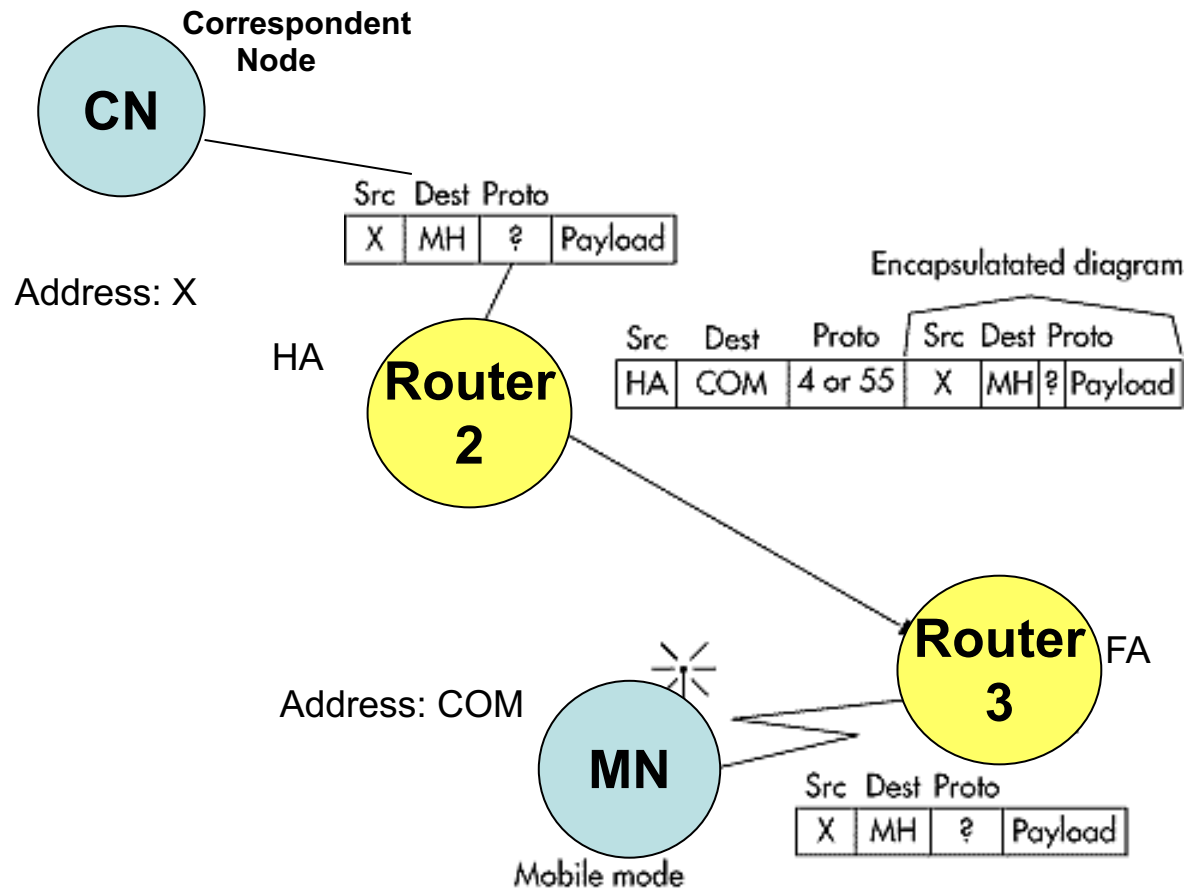
Mobile IP binding



Tunneling in Mobile IP

- When CN sends the data to MN, it uses the original address of the MN, so the packet goes to HA.
- From the mobility binding HA encapsulates the packet (IP-in-IP or GRE) and sends to CoA.
- The default encapsulation mechanism that must be supported by all mobility agents is IP-within-IP (RFC 2003)
- Using IP-within-IP, home agent inserts a new IP header in front of the IP header of any datagram
- The FA de-capsulate the packet and extracts the original packet that was sent by the CN.
- The FA then sends this packet to the MN using the Home address destination.
- The reverse route from MN to CN may or may not follow this path. We have two options
 - **Triangle routing – Reply pkts are sent directly to CN from MN**
 - **Reverse Tunneling – Reply pkts are tunneled to HA by FA.**

IP-in-IP Illustration



- Destination address set to the care-of address assigned to the MN (COM)
- Source address set to the home agent's address
- Tunnel header uses 4 for higher protocol id – this ensures that IP after stripping out the first header, processes the packet again
- Tunnel header of 55 used if IP minimal encapsulation used (RFC 2004)

Problems facing Mobile IP

- Routing inefficiencies
 - Asymmetry in routing: Triangle routing
 - Route optimization requires changes in the correspondent nodes that will take a long time to deploy
- Security issues
 - Firewalls
 - Blocks all classes of incoming packets that do not meet specified criteria
 - It presents difficulties for mobile nodes wishing to communicate with other nodes within their home enterprise networks

Problems facing Mobile IP (cont)

- Security issues
 - Ingress filtering
 - Many border router discard packets coming from within the enterprise if the packets do not contain a source IP address configured for one of the enterprise's internal network
 - Mobile node would otherwise use their home address as the source IP address of the packets they transmit
 - Possible solution: tunneling outgoing packets from the care-of address (Q: where is the target for the tunneled packets from the mobile node? Home agent?)

Mobile IP: Other Issues

- **Reverse Tunneling**

- firewalls permit only “topological correct” addresses
- a packet from the MN encapsulated by the FA is now topological correct

- **Optimizations**

- Triangular Routing
 - HA informs sender the current location of MN
- Change of FA
 - new FA informs old FA to avoid packet loss, old FA now forwards remaining packets to new FA